Emmanuel Amissah Sampson

Urban Analytics • Urban Mobility • Spatial Data Science • Sustainable Cities

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RESEARCH PROFILE

Master of Urban Spatial Analytics candidate at the University of Pennsylvania with graduate training in development management and business (LSE MSc; Lancaster MBA, Distinction), and roughly a decade of geospatial industry experience across more than thirty emerging markets. Research examines how urban form and planning shape the geography of everyday commercial destinations, how openings, closures, and relocations reshape accessibility, pedestrian activity, and sustainable mobility, and who gains and who bears the costs when private location decisions and public planning goals diverge.

RESEARCH INTERESTS

Urban analytics and spatial data science • Urban mobility, walking, and pedestrian activity • Commercial location and the geography of everyday destinations • Network accessibility and urban form • Land use, zoning, and planning • Sustainable travel and transport emissions • Distributional equity and spatial data gaps • Causal inference with spatial spillovers • Comparative urbanization in African cities

EDUCATION

Master of Urban Spatial Analytics (MUSA) • Expected May 2027 • STEM-Designated
University of Pennsylvania • Philadelphia, PA, USA • Stuart Weitzman Scholar
Fall 2026 coursework: Statistical and Data Mining Methods for Urban Data Analysis; Public Policy Analytics; Geospatial Data Science in Python; AI for Urban Sustainability; Financing Climate-Resilient City Infrastructure.
Planned, spring 2027: Introduction to Causal Inference; Deep Learning with Python; Geospatial Cloud Computing.

Master of Business Administration (MBA), Distinction • 2019
Lancaster University • Lancaster, UK
Coursework: Business Economics; Operations Management; Business Research and Consultancy; Accounting and Finance for Managers.

MSc Development Management • 2015
London School of Economics and Political Science • London, UK
Academic focus: Economic Development Policy; Quantitative Analysis; Research Design and Dissertation.

BSc Business Administration (Banking & Finance), First Class Honours • 2009
Kwame Nkrumah University of Science and Technology • Kumasi, Ghana
Quantitative foundation: Business Mathematics; Business Statistics; Quantitative Methods; Business Economics; Introduction to Operations Management.

SELECTED PUBLICATIONS

Sampson, E. (2026). Ethical AI in Financial Systems: A Risked-Based Framework for Responsible Innovation. International Journal of Management and Organizational Research, 5(1), 104–111. doi:10.54660/IJMOR.2026.5.1.104-111.

Sampson, E. (2025). AI, Ethics, and Access: Designing Inclusive Digital Systems for the Next Billion Users. IIARD International Journal of Economics and Business Management, 11(9), 493–513. doi:10.56201/ijebm.vol.11.no9.2025.pg493.513.

Sampson, E. (2025). AI and Financial Inclusion: Analyzing the Impact of Intelligent Systems on Underserved Populations. International Journal of Financial Management and Economics, 8(1), 606–619. doi:10.33545/26179210.2025.v8.i1.700.

Sampson, E., & Narteh-Kofi, E. (2025). Digital Sales Transformation in Sub-Saharan Africa: Impacts on Cross-Border Trade and Global Market Integration. World Journal of Advanced Research and Reviews, 27(3), 1092–1101. doi:10.30574/wjarr.2025.27.3.3249.

Ogundipe, F., Sampson, E., Bakare, O. I., Oketola, O., & Yusuf, R. A. (2022). Technology for a Sustainable Future: Unlocking the Power of Digital Transformation. Journal of Frontiers in Multidisciplinary Research, 3(2), 146–156. doi:10.54660/.IJFMR.2022.3.2.146-156.

RESEARCH EXPERIENCE

Graduate Research and Coursework • University of Pennsylvania • 2026–Present

- Public Policy Analytics (MUSA 5080): built a cross-validated model of car-free households across 381 Philadelphia tracts, engineering spatial features from city open-data APIs (crime, 311, bike share) and cutting prediction error by 25%.
- Statistical and Data Mining Methods (MUSA 5000): modeled median house values across 1,720 Philadelphia block groups with multiple regression in R, using log transformations, stepwise AIC selection, five-fold cross-validation, and residual diagnostics that identified spatial clustering in errors (team of three).

- Geospatial Data Science in Python (MUSA 5500): analyzed Philadelphia evictions and code violations with GeoPandas and NumPy; built a flood-inundation workflow from digital elevation models with Rasterio, Xarray, and GeoPandas.
- AI for Urban Sustainability (MUSA 6950): computed NDVI and vegetation cover for 207 Boston census tracts from NAIP aerial imagery via the Planetary Computer STAC API; wrote PyCUDA GPU kernels to model slope and hourly building shadows from a digital surface model.
- Trained in spatial autocorrelation, spatial regression (spatial lag, spatial error, geographically weighted regression), logistic and multinomial choice models, point pattern analysis, and clustering.
- Developing a research agenda on commercial location, accessibility, and sustainable mobility: how urban form, zoning, and pedestrian networks shape where everyday destinations emerge, and how openings, closures, and relocations, including the 2023–2025 Rite Aid closures, redistribute pedestrian activity, access, and travel, using establishment records, mobility data, and quasi-experimental designs with spatial spillovers.

TECHNICAL AND QUANTITATIVE PREPARATION

- Programming and analytics: Python (pandas, GeoPandas, NumPy, Rasterio, Xarray, PyCUDA, Matplotlib); R (tidyverse, tidycensus, caret, lmtest); statistical and data-mining methods; reproducible analysis.
- Spatial methods: GIS (ArcGIS Pro, QGIS, GeoDa); spatial autocorrelation and spatial regression; network-based and spatial interaction accessibility analysis; raster/vector analysis; gravity/Huff models.
- Developing: causal inference (event studies, difference-in-differences, spatial interference), pedestrian network analysis, firm location-choice and establishment survival models, and machine learning.

PROFESSIONAL EXPERIENCE

Leica Geosystems (Hexagon) • Geospatial Regional Manager, East and Central Africa • Feb. 2025–Aug. 2026

- Led market strategy, P&L, partnerships, and cross-functional delivery for geospatial and construction-technology solutions; delivered $10.2 million in regional revenue across mining, government, and construction accounts.
- Managed a $2.2 million operating budget and 23 employees, including equipment inventory, logistics, and distributor and vendor relationships.

Vela Sustainability Technologies • Business Development Consultant, Western and Central Africa • Sept. 2024–Jan. 2025

- Supported market development for industrial-hygiene and occupational-health software and engaged stakeholders on exposure monitoring, compliance, and sustainability performance.

Trimble Inc. • Geospatial Business Development Manager, Market Expansion and Strategic Accounts • Sept. 2015–Dec. 2023

- Led institutional and funded-project development across African markets; developed donor intelligence and engagement strategies contributing to an approximately $85 million funded-project pipeline over eight years.
- Helped secure land and site permits for continuously operating reference station (CORS) networks supporting land administration projects across emerging markets.
- Supported a $3 million World Bank-funded project in Tanzania; worked with public agencies, consultants, and technical teams to translate spatial data into infrastructure and public-sector decisions.
- Supported adoption of GIS, GPS/GNSS, surveying, mapping, reality capture, and digital-twin technologies across mining, construction, infrastructure, and government settings.

De La Rue • Strategic Account Manager, West and Central Africa • May 2021–Oct. 2022

- Managed strategic public-sector and enterprise accounts, including Ghana's Digital Tax Stamp solution.

Catholic Relief Services / CARE International / AngloGold Ashanti Malaria Control / MTN Ghana • Ghana and Côte d'Ivoire • 2009–2015

- Co-led value-chain research and advisory work on the Cocoa Life program, examining constraints and opportunities facing farmer-based organizations.
- Developed procurement and logistics strategy for a five-year, $150 million Global Fund malaria-control project.
- Led continuous-improvement work that reduced inventory days by 40% and generated approximately $1 million in savings.

FELLOWSHIPS, AWARDS AND SCHOLARSHIPS

- Stuart Weitzman Scholar, Stuart Weitzman School of Design, University of Pennsylvania — 2026
- Fellow, Institute of Management Consultants, Nigeria — 2026
- Perplexity AI Business Fellowship — 2025
- Staple Trust Bursary, London School of Economics and Political Science — 2013
- Ambassador E. O. Kolade Scholarship, London School of Economics and Political Science — 2013

SERVICE AND ACTIVITIES

- Mentor to students in Ghana pursuing scholarship-funded graduate study at U.S. universities.
- Member, Student Planning Association (MUSA), Stuart Weitzman School of Design Student Association, and Penn African Students Association, University of Pennsylvania.